

Changes to Atmospheric and Oceanic Fluid Dynamics for Second Printing

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June 6, 2007

The changes throughout are small and only in the details. No new sections, subsections or figure have been added. The page count and page numbering is the same - all chapters start on the same page numbers etc - and the equation numbering is the same.

1. Page 5, eq. (1.10). Change b_x to b^x .
2. Page 29, equation on last line. dp should be $d\alpha$ in middle term on rhs.
3. Page 31, line below (1.139). Should be: $\theta = \theta(\eta, S, p_R) = \theta(T, S, p)$; i.e., p not p_R in last term.
4. Page 35, Fig. 1.4 caption, line 3. Replace 13.36°C by 10.14°C .
5. Page 86, line 3. The factor of $\sin \theta$ should be removed.
6. Page 119, problem 2.14. The pressure term, $-\rho_0^{-1}\nabla p$ added to right-hand side of the first equation.
7. Page 150, (3.149) and (3.150). Factor of $e^{i(kx+ly)}$ added.
8. Page 156, eq. (3.178). Add minus sign. Eq. (3.179). Factor of $1/2$ added, and the minus sign deleted. The factor of $1/2$ added to (3.183).
9. Page 160. Formatting error in problem 3.3. It should read $\nabla \cdot [\mathbf{v}(E + gh^2/2)]$ and not $\nabla_{\mathbf{v}(E+gh^2/2)} \cdot$.
10. Page 185, eq. (4.109). Should read $\alpha^{-2} \gg 1$ and not $\alpha^2 \ll 1$. Also, line before (4.113), $\Omega = f\mathbf{k}$ should be $2\Omega = f\mathbf{k}$.
11. Page 196, eq. (5.1). Should read $(u, v) \sim U$, not $(u, v) \sim L$.
12. Page 208, eq. (5.61). Missing subscript 0 on second $\hat{\mathbf{u}}$, and second t should be $\hat{t}$. Also, missing $\hat{f}_0$ in (5.62), although its value is unity.
13. Page 213, eqs. (5.85a) and (5.88a). Add minus sign, change H_2 to H_1 in last term in (5.88a).
14. Page 227, eq. (5.159). H should be H^2 .
15. Page 228, eq. (5.168). As written, $\hat{\psi} \equiv (\psi_1 - \psi_2)/\sqrt{2}$. Also, in (5.170), $\int d\mathbf{x} \equiv \int dA$.
16. Page 239, eq. (5.225). Last term should have minus sign before it.
17. Page 256, caption to Fig. 6.5. $\sigma = -ikc_i$ not $\sigma = kc_i$.
18. Page 259 eq. (6.53) and (6.54). The terms $v\zeta$ should be multiplied by 2. Eq. (6.55), the two plus signs should both be minus signs, as in (7.17).

19. Page 262, Fig. 6.9. The labels B and C are interchanged.
20. Page 263, eq. (6.61). z_A instead of z_B in fourth term in second expression.
21. Page 265, after (6.74). Should be ‘the integral must vanish’, or alternatively (and implying) ‘the integrand must vanish at some point’.
22. Page 266. Above (6.78), change to 0 to L from $\pm L/2$.
23. Page 274, l-2. $\beta \neq 0$, not $\beta = 0$.
24. Page 277, surrounding (6.128). K instead of k , so the unstable interval is $\sqrt{\beta/2U} < K < k_d$.
25. Page 275, first sentence of second section. No hats on variables.
26. Page 279. Better to have primes on the perturbation variables in (6.135)–(6.137).
27. Page 280, eq. (6.141). c_T and c_B are interchanged.
28. Page 284, eq. (6.161) should read: $(\partial/\partial t + U\partial/\partial x)q' + (\partial\psi'/\partial x)(\partial Q/\partial y) = 0$ and not $(\partial/\partial t + U\partial u/\partial x)q' + (\partial\psi'/\partial x)(\partial Q/\partial y) = 0$. (Spot the difference.)
29. Page 285, eq. (6.165). ψ should be ψ' on lhs, and four lines above.
30. Page 296. Just before (7.5), J changed to H in a few spots.
31. Page 298, first sentence of section 7.2. ‘Eddy flux of potential may be’ should be ‘Eddy flux of potential vorticity may be’.
32. Page 300, eq. (7.33). k and l are transposed in the expressions for $\tilde{u}$ and $\tilde{v}$. We should have: $\tilde{u} = -il\tilde{\psi}$, $\tilde{v} = ik\tilde{\psi}$.
33. Page 303, l 7. p should be m , twice.
34. Page 304, eq. (7.60b). w should be $\bar{w}$.
35. Page 308. In (7.79), the first $\bar{\sigma}$ should be $\bar{h}$. Better, $\bar{v}^b \equiv \bar{v}\bar{\sigma}^b/\bar{\sigma} = \bar{v}^b + \bar{v}'\bar{\sigma}'^b/\bar{\sigma}$.
36. Page 310, second bullet. E should be $\mathcal{F}$.
37. Page 311, eq. (7.93). ϕ in first term should be $\bar{\varphi}$.
38. Page 313, eq. (7.106). Should be $F[\varphi]$ not $F[\bar{\varphi}]$.
39. Page 314, eq. (7.112c). Middle equation should have minus sign on rhs, and in last equation $\mathbf{v}'$ should be $\mathbf{u}'$.
40. Page 319, eqs. (7.130) and (7.131). Missing factor of ρ_0 , and, in (7.131) only, f_0 .
41. Page 320, just above (7.138). Reference to (7.136) should be to (7.135).
42. Page 324, eq. (7.141). Subscripts 1 should be subscripts i .
43. Page 327, eq. (7.157). No minus sign on rhs.
44. Page 330, first equation in box (7.812). Extraneous s on lhs, next to $\hat{H}$.
45. Page 339, eq. (8.8). Should be ζ and not ψ in viscous term.
46. Page 340. $1/L^2$ should be L^2 in (8.11), and k^4 should be k^2 in (8.12).
47. Page 345. In fig. 8.3, $E(k)$ should be $\mathcal{E}(k)$.

48. Page 352 and 353, eq. (8.48) and eq. (8.53). $\hat{E}$ should be $\bar{E}$, and $\hat{Z}$ should be $\bar{Z}$, respectively.
49. Page 352, eq. (8.49). $Z(q)$ is not the same function as the $Z(k)$ of (8.44), but it is still such that the enstrophy is $\int Z(q) dq$.
50. Page 355, eq. (8.62a). In (8.63)–(8.65), change $\mathcal{E}(k)$ and not $E(k)$.
51. Page 369, eq. (8.105). χ should be χ' on right-hand side.
52. Page 380, caption and inset to Fig. 9.1. ε should be $\varepsilon^{1/3}$.
53. Page 386, eq. (9.24b). a and b are transposed on second term on rhs of definition of Jacobian. Also, line before (9.27), second ‘barotropic’ should be ‘baroclinic’.
54. Page 389. 1st text line should read $|\nabla^2| \sim k^2 \ll k_d^2$ and not $\nabla^2 \sim k_d^2 \ll k^2$.
55. Page 392, eq. (9.48). $D[\tau]$ is multiplied by $-k_d^{-2}$.
56. Page 399, fig. 9.9 caption. Change in wording to correct error.
57. Pages 426–436. ‘Across gradient’ changed to ‘across iso-surfaces’ and ‘diagradient’ replaced by ‘along-gradient’ in a few places.
58. Page 428, line after (10.100). ‘Divergence’ should be ‘gradient’, twice.
59. Page 434, eq. (10.132): $\bar{u}$ should be $\bar{\mathbf{u}}$ in first term. Minor change to sentence following.
60. Page 437, eq. (10.146). Subscripts on s should be superscripts, as they denote components and not derivatives.
61. Page 440, eq. (10.156). The $\mathbf{u}'$ should be outside of the derivative in the third order term, but this term is promptly neglected.
62. Page 443, eq. (10.175). $\overline{\mathbf{v}'\sigma'}$ should be $\overline{\mathbf{u}'\sigma'}$ in third term.
63. Page 458 eq. (11.4). Need a factor of $1/a$ on first term of rhs. Also, ∂_y should be ∂_{ϑ} on the following line; alternatively, delete factor of a^{-1} .
64. Page 460, eq. (11.11). Missing factor $-1/g$ on rhs.
65. Page 468, item (ii) and (11.40). Note that we are using an *equatorial* beta plane, on which $f_0 = 0$. Thus, $f = f_0 + \beta y = \beta y$.
66. Page 471, eq. (11.54). $\delta\theta$ should be θ .
67. Page 471. Line above (11.57), should be $m(0) = \Omega a^2 \cos^2 \vartheta$ (i.e., ϑ and not ϑ_1), as $u = 0$ at the surface. And (11.57) and (11.58) are missing factors of $-\theta_0$ on rhs.
68. Page 474, $\cos \vartheta$ should be $\cos^2 \vartheta$ in text at end of second paragraph.
69. Page 476, eq. (11.64); page 480, eq. (11.73); page 481, eq. (11.75). The eddy flux term needs a factor of $1/a$.
70. Page 494, eq. (12.34). Factor of $1/2$ added to right-hand sides. See, e.g., (12.27).
71. Page 504. Line after (12.61b), extraneous $-\nabla$ between g' and η .
72. Page 505, eq. (12.71). h_i should be $\bar{h}_i$.
73. Page 506, eq. (12.77). v_1 on rhs should be v_2 . (Eq. (12.79) is correct.)
74. Page 507. Eq. (12.83), second term on rhs should have plus sign, not minus sign.

75. Page 508. Eq. (12.85), q should be $\bar{q}$, twice.
76. Page 511. Sentence above (12.89), polewards and equatorwards are reversed. And ‘inequality’ should be ‘equality’ in the line following (12.89).
77. Page 513. Minor change to (12.95) and (12.96).
78. Page 514. In (12.98) η should be $\bar{\eta}$, and in (12.99) q should be $\bar{q}$.
79. Page 516. In (12.103), factor of 1/2 should be inside square brackets.
80. Page 518. Line 2: decreases should be increases. Line 7, westward should be westerly.
81. Page 531, fifth line after (12.119). Tropopause should be troposphere. Likewise on page 533, third line after (12.123)
82. Page 533, fourth line after (12.120), second subscript should be s , not i .
83. Page 535, eq. (PV.1). The term $-\partial \bar{v}_i \bar{q}_i / \partial y$ changed to $\bar{v}'_i \bar{q}'_i$.
84. Page 539, endnote 12. The reference to page 12.6.3 should be to page 532, or to section 12.6.3.
85. Page 543. Missing prime on v in (13.7) and (13.10).
86. Page 547. The factor $1/k_s$ should be omitted in (13.18).
87. Page 551, below (13.34). Would be clearer if it read: ‘... and so *short* waves are trapped by westerly flow.’
88. Page 555, eq. (13.50), and later in (13.66). First term, $\partial \psi / \partial t$ should be $\partial (\partial_z \psi) / \partial t$.
89. Page 556. In (13.60a), minus sign after $\bar{u}$ should be plus sign [as in (5.186b)].
90. Page 555–559. h_b replaced by $\tilde{h}_b$.
91. Page 558. In (13.65), im should be $-im$. In the paragraph following, it is $\rho_R |\tilde{\psi}|^2$, not the amplitude of $\tilde{\psi}$ itself, that is independent of height in the last case.
92. Page 574. The right-hand sides of (13.93) and (13.94) should be multiplied by -1 .
93. Page 586, eqs. (14.5) and (14.14). The quantity W_E changed to F_τ in a few places.
94. Page 725. Holloway and Hendershott (1986). Year should be (1977).